Tien and Tieu *Discover Computing* (2026) 29:436
https://doi.org/10.1007/s10791-026-10319-2

Discover Computing

RESEARCH

Open Access

Synergistic multi-stage processing for enhanced context relevance in RAG architecture

Luyen Nguyen Tien[1] and Binh Hoang Tieu[1*]

*Correspondence:
Binh Hoang Tieu
htbinh@cmcu.edu.vn
[1]Faculty of Information and Communication Technology, CMC University, Hanoi, Vietnam

Abstract

In recent years, Retrieval-Augmented Generation (RAG) systems have faced critical limitations in semantic consistency, cross-model compatibility, and knowledge-aware response generation. This study introduces SynRAG, an integrated framework designed to overcome these challenges by combining four main innovations: hierarchical semantic segmentation supported by clustering methods, a multi-phase hybrid retrieval mechanism, embedding dimension adjustment strategies, and context-aware prompt engineering. Experimental evaluation on NarrativeQA, QASPER, and QuALITY demonstrates superior performance: on NarrativeQA, the framework achieves 95.3% faithfulness and 96.1% answer relevancy, representing + 11.1 pp and + 21.0 pp improvements respectively; across other datasets, consistent gains are observed on grounding, retrieval (NDCG), and generation metrics. Notably, our dimension transformation maintains high retention relative to 1536-dim embeddings: 95.1% Faithfulness, 94.5% Answer Relevancy, and 93.1% NDCG on average when reduced to 768-dim. Comprehensive evaluation demonstrates consistent improvements across all key metrics, highlighting the robustness of the RAG system. SynRAG marks a notable step forward in RAG development by offering an integrated framework that effectively connects research-level innovations with real-world deployment needs, ensuring both efficiency and scalability in computation.

Keywords RAG, Semantic chunking, Hybrid search, Prompt engineering, Natural language processing

1 Introduction

Retrieval-Augmented Generation (RAG) has become a pivotal approach that connects information retrieval techniques with large language models (LLMs), enabling systems to generate context-aware and knowledge-grounded responses by utilizing external data sources [32]. Over time, RAG methodologies have evolved substantially–from early keyword-matching systems to advanced neural architectures that employ deep learning for semantic representation and understanding [7, 52]. This evolution has resulted in major improvements in response accuracy and contextual relevance, especially in areas that demand recent or specialized knowledge beyond the static limits of closed-book language models.

Nevertheless, as RAG architectures have become increasingly complex and heterogeneous, they now face four major obstacles that constrain their real-world performance. The first concerns the way documents are divided into semantically coherent segments. Many existing systems still rely on rigid or rule-driven segmentation strategies, such as fixed-length windows, which handle text in a purely mechanical manner. Recent research demonstrates that context-aware semantic chunking can enhance retrieval accuracy by 15–25% [13, 14, 65]. Despite this progress, most frameworks continue to adopt oversimplified segmentation techniques that scatter essential content across multiple chunks, resulting in information loss and diminished contextual fidelity [28, 53]. To overcome this, more adaptive chunking mechanisms are needed–ones capable of adjusting to diverse document layouts while maintaining overall semantic consistency.

Second, the growing diversity of embedding models, which vary widely in dimensionality (from 384 up to 4096), introduces major interoperability issues [6, 18]. These discrepancies hinder the smooth alignment and joint utilization of heterogeneous embeddings within a unified RAG workflow. The third challenge pertains to the retrieval strategy. A large portion of current RAG systems still depend on a single retrieval paradigm–either vector-based semantic search or keyword matching–thereby missing the potential synergy between the two approaches [12, 17]. While semantic search effectively captures conceptual or contextual relationships, keyword retrieval remains more precise for identifying named entities and infrequent terms. The question is: how can we effectively combine these approaches without creating computational overhead or introducing redundancy? The fourth challenge lies in how we structure our prompts and manage conversation context. Most current systems use flat, static prompting strategies that don't fully utilize the rich metadata available in knowledge bases or the conversational history that could provide valuable context [15, 57]. This leads to issues with attribution, grounding, and the overall quality of generated responses. These four challenges–intelligent semantic chunking, embedding interoperability, effective hybrid search, and knowledge-aware prompting–represent the core bottlenecks that limit RAG systems' feasibility across diverse domains. Addressing them requires a holistic approach that considers the entire pipeline, from document processing to response generation.

In this paper, we present SynRAG, a comprehensive framework designed to tackle these challenges through four integrated improvements. Our journey begins with a fundamental reimagining of document processing through multi-layer semantic chunking. Rather than treating chunking as a simple segmentation step, we employ an adaptive framework that utilizes HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) [38] and Agglomerative clustering [56] , together with a refined mechanism for outlier detection and correction. This approach dynamically adjusts to different document structures while maintaining semantic integrity, achieving strong performance (88.7–95.3% faithfulness and 90.1–96.1% answer relevancy) across our evaluation datasets.

Building upon this foundation, we develop a sophisticated multi-stage hybrid search architecture that combines dense vector retrieval with POS/LLM-driven (Part-Of-Speech/Large Language Model) keyword extraction, implementing duplicate-aware fusion and neural reranking. This approach leverages the complementary strengths of both semantic and keyword-based methods, achieving significant improvements in recall and precision while maintaining computational efficiency [30]. To address the diversity of embedding models with varying dimensions (384–4096), we develop versatile embedding alignment methods, combining DCT-based upsampling, linear projection, weighted redistribution, and PCA-based compression (PCA-Principal Component Analysis). These approaches allow embeddings of different dimensions to interact effectively, preserving semantic integrity and enhancing cross-model compatibility.

Finally, we complete our framework with knowledge-aware prompt engineering that integrates system instructions, knowledge base metadata, and conversation memory through structured JSON templates. This approach improves grounding and attribution while demonstrating significant improvements in faithfulness and answer relevancy compared to traditional methods. Unlike recent 'synthetic-RAG' approaches that attempt to distill knowledge from generated data–which risks amplifying model biases–SynRAG or Synergistic RAG focuses on maximizing Signal Fidelity from authentic sources. Our contribution is establishing a pipeline where semantic chunking and dimension transformation work synergistically to preserve the integrity of human-authored content across domains. By integrating these four improvement into a cohesive framework, we've proposed a RAG system that addresses the fundamental challenges facing the field today by integrating all key components of RAG into a cohesive framework.

The paper is structured as follows: Sect. 2 reviews existing literature on semantic chunking, hybrid retrieval, embedding alignment, and prompt optimization. Section 3 outlines our methodology, detailing the multi-level semantic chunking process, hybrid retrieval architecture, embedding dimension alignment, and context-aware prompt construction. Section 4 presents experimental findings, including dataset benchmarks, ablation studies, and comparative analyses. Section 5 discusses the broader implications of our results and contrasts them with prior work, while Sect. 6 concludes with potential avenues for future exploration and real-world applications.

2 Literature review

2.1 RAG across multiple knowledge domains

The evolution of RAG systems has been marked by significant advances across multiple dimensions, each addressing critical bottlenecks in information retrieval and generation. RAG systems have demonstrated significant value across diverse industries, addressing unique domain-specific challenges through specialized implementations. In healthcare, systems like MKRAG have boosted the capabilities of medical Q&A by connecting to specialized, external knowledge sources, while recent work has prioritized topical relevance and factual accuracy in health information retrieval [49, 51]. The legal domain has benefited from hybrid approaches combining RAG with Case-Based Reasoning (CBR), improving legal research efficiency through enhanced retrieval of relevant precedents and statutes [58]. Additionally, systems like LegalAsst have streamlined court procedures through AI-powered legal case retrieval [16]. In education, RAG systems have shown promise in mathematics education by drawing upon validated information from trusted academic sources, though designers must carefully balance educational alignment with student preferences to maintain effective learning outcomes [31]. It is important that RAG systems have a strong impact on different knowledge domains, but the requirements for reliability and how to communicate with LLM to accurately extract this knowledge are limited.

2.2 Advancements in RAG technology

Researchers have developed numerous advanced RAG variants to overcome these shortcomings and enhance the conventional framework. This section summarizes the state of the art of previous research (Table 1) and the extent to which they have been deployed in key areas that we believe are critical to achieving a comprehensive RAG system.

2.2.1 Chunking and document processing

The foundation of any effective RAG system lies in how documents are segmented and processed. Traditional approaches have relied on fixed-size tokenization methods [32] and rule-based segmentation strategies [12, 35], which, while functional at a basic level, frequently break key information apart. This fragmentation causes context to be lost and ultimately degrades the effectiveness of retrieval.Recent studies have demonstrated that intelligent semantic chunking can improve retrieval precision by 15–25% [13, 14]. Clustering-based approaches have emerged as promising alternatives, with HDBSCAN [38] and Agglomerative clustering [56] showing particular promise for maintaining semantic boundaries. However, existing implementations often struggle with outlier handling and fail to adapt to diverse document structures. Wilson et al. explored clustering-based chunking but noted limitations in maintaining semantic coherence across different document types. This gap motivates our multi-layer semantic chunking approach that combines advanced clustering algorithms with intelligent outlier handling strategies.

2.2.2 Search architectures

The retrieval landscape has evolved from single-modality approaches to sophisticated hybrid architectures that leverage the complementary strengths of different search paradigms. Classical work combined multiple evidence sources via linear weighting or voting mechanisms [29]. Modern dense-sparse hybrids exploit complementary strengths: dense encoders yield paraphrase-robust semantics through models like DPR [23], Contriever [19], and ColBERT [25], while sparse methods such as BM25 [46] and SPLADE [10] excel at recovering rare terms, identifiers, and entities [12]. Neural rerankers, particularly cross-encoders and BGE models (BAAI General Embedding) [2], have shown significant improvements in result ordering, though at the cost of additional latency. Yet many existing hybrid systems remain single-pass, fusing scores without staged expansion, deduplication, or cost-aware batching, which can hurt recall-precision

Table 1 Previous works addressing key components in RAG enhancement

Authors	Component	Key technique
[45]	Chunking	Semantic Chunking
[33]	Chunking	Clustering-based
[23]	Search	Dense Retrieval
[25]	Search	ColBERT
[40]	Search	Neural Reranking
[57]	Prompt	Chain-of-Thought
[42]	Prompt	Structured Prompt
[50]	Domain	Medical MKRAG
Our work	Overall	SynRAG

trade-offs and propagate redundancy. Our approach addresses these limitations through a four-stage pipeline that includes semantic retrieval, POS/LLM-driven keywording, duplicate-aware fusion, and neural reranking, executed with batching and caching to balance quality and computational cost.

2.2.3 Embedding and interoperability

The rapid growth of embedding models, which feature a wide range of dimensions (from 384 to 4096), has led to significant interoperability issues within RAG systems [6, 18]. Traditional approaches often require retraining or fine-tuning when switching between different embedding architectures, limiting the flexibility of RAG systems to leverage the best model for each component. Recent work has explored various transformation techniques, including linear projection methods and dimensionality reduction approaches. Khaledian et al [24] has highlighted the potential of dimension reduction not only for compatibility but also for performance optimization. PCA-RAG demonstrates that applying Principal Component Analysis (PCA) to compress high-dimensional embeddings can yield substantial improvements in retrieval speed and memory efficiency, with minimal impact on semantic fidelity. However, most existing solutions focus on specific model pairs or require extensive retraining. The central challenge is to create transformation methods capable of maintaining semantic integrity while facilitating effortless integration across different models. Our approach utilizes a set of comprehensive techniques for dimension transformation–namely, DCT-based upsampling, linear projection, weighted redistribution, and PCA reduction. The goal is to resolve these compatibility issues without compromising the semantic accuracy.

2.2.4 Prompt engineering

One of the final steps before sending to LLM to generate answers is how to restructure the contextual components into structured prompts. Prompt engineering has evolved from simple template-based approaches to sophisticated systems that integrate contextual information and conversation history. Template prompts and instruction tuning are widely used [57], while dynamic and chained prompting have shown improvements in multi-step reasoning tasks [55, 62]. However, knowledge-base-aware structuring–explicit fields for system instruction, KB metadata, top-k context with document names, user query, instruction template, and conversation memory–remains underexplored for grounding and attribution at inference time. Flat concatenation approaches blur field boundaries, increase data leakage, and inflate token usage. Recent work by Gupta et al. [15] has highlighted the importance of structured prompting for improving grounding and attribution in RAG systems. Our approach contributes a JSON-style schema that maintains role separation, supports principled token budgeting, and empirically improves faithfulness and answer relevancy compared to traditional flat concatenation methods.

Multi-stage RAG systems have shown steady enhancements over single-modal processing [12, 32]. On the other hand, recent surveys and system papers have emphasized the importance of concurrency, caching, and vector storage (e.g., FAISS [8], pgvector [44]) to achieve scale [22]. However, most existing systems focus on individual components rather than integrated frameworks. The challenge lies in developing systems that can effectively combine multiple improvement while maintaining computational efficiency and implementation flexibility. This integration challenge becomes even more complex when considering the interdependencies between different components, such as how chunking strategies affect retrieval performance, or how embedding transformations impact the effectiveness of hybrid search.

2.2.5 Feature enrichment vs. retrieval reliance

Beyond text-specific retrieval, recent progress in Visual Question Answering (VQA) offers valuable theoretical insights for evaluating context relevance in RAG systems. Robustness under distribution shift has been analyzed in R-VQA [5], revealing the vulnerability of standard retrieval models to adversarial noise and out-of-distribution data. This underscores the need for strong feature representation, a principle embodied in SynRAG's multi-stage hybrid retrieval. The feature enrichment approach in ENVQA [4] shows that augmenting inputs with structured metadata significantly improves reasoning compared to plain retrieval. Similar ideas are explored in recent RAG literature, including hybrid KG-vector enrichment for domain-centric Q&A [54] and retrieval-augmented feature generation for domain-specific tasks [64]. Hallucination risks arising from parametric language priors are examined in ESC-Net [3]. SynRAG addresses this by enforcing strict evidence grounding through multi-stage fusion and neural reranking, in line with advances such as Stable-RAG [63] and MEGA-RAG [60], which reduce permutation-induced and multi-evidence-related hallucinations respectively. These connections motivate SynRAG's integrated design, combining hybrid retrieval, structured prompting, and adaptive chunking to strengthen robustness, context enrichment, and evidence fidelity in text-based RAG.

Finally, despite significant improvements, previous studies have focused on individual domains without a comprehensive synthesis or study of the overall combination of methods. For a RAG system to perform comprehensively in terms of accuracy and practical performance, the above factors are key points to be able to influence. In this study, we propose a comprehensive SynRAG system by tightly combining the above areas from semantic segmentation and chunking to staged hybrid retrieval with neural re-ranking and structured reminder techniques, and provide system controls that clarify the trade-off between quality and cost in production deployment. Subsequent experiments showed some significant improvements over the above studies.

3 Methodology

The SynRAG comprehensive system operates through a sophisticated multi-stage pipeline that seamlessly integrates document processing, retrieval, and generation components. The system architecture is designed to handle the complexity of real-world RAG applications while maintaining computational efficiency and scalability, as detailed in Fig. 1.

The system begins with intelligent document processing through multi-layer semantic chunking, which preserves semantic coherence while adapting to diverse document structures. This approach addresses the limitations of traditional fixed-size chunking methods) by employing advanced clustering algorithms that maintain semantic boundaries. The processed content is then transformed through comprehensive embedding dimension transformation techniques, enabling seamless integration of diverse embedding models with varying dimensions (384–4096) [18]. The retrieval stage employs a

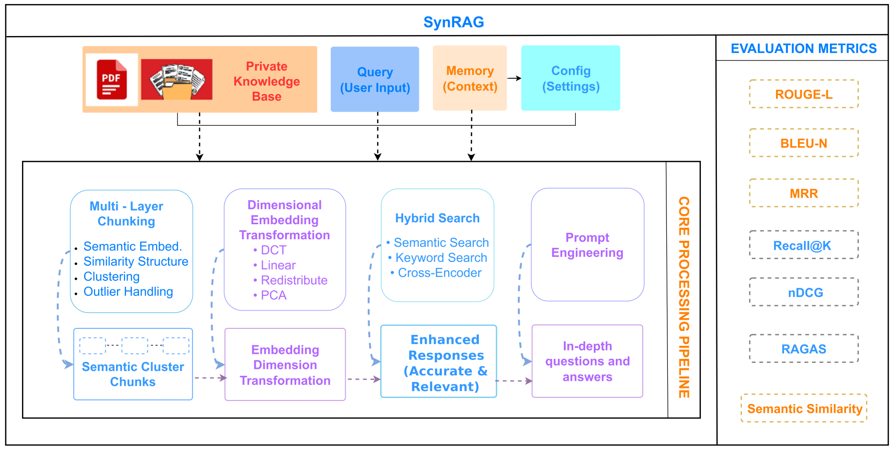

Fig. 1 Overview of the SynRAG comprehensive system highlighting four core modules: multi-layer chunking, embedding-dimension transformation, hybrid search, and knowledge-aware prompt engineering

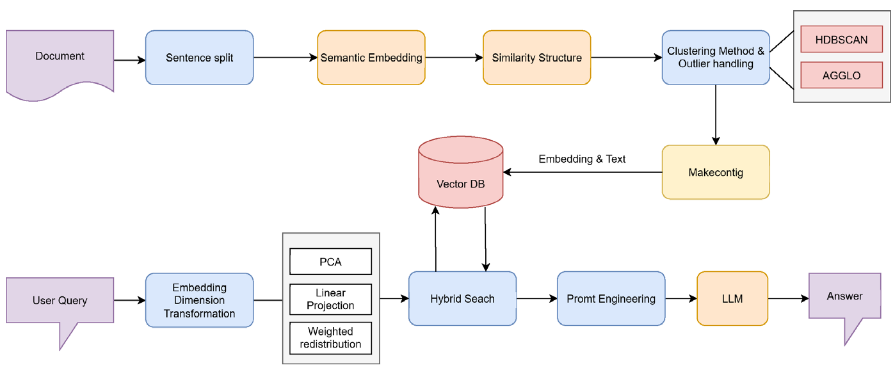

Fig. 2 An overview of the two-pipeline framework, where document processing (top) is handled by the Semantic Cluster Chunking method (configurable clustering and chunking), while query processing (bottom) is managed via hybrid search and LLM-based answer generation. The system's enhanced retrieval performance stems from the integration of multi-layer semantic chunking with embedding dimension transformation

sophisticated multi-stage hybrid search architecture that combines semantic and keyword-based approaches with intelligent fusion and neural reranking, leveraging the complementary strengths of dense and sparse retrieval methods [23, 46]. Finally, the generation stage utilizes knowledge-aware prompt engineering to structure contextual information and conversation memory for optimal response generation, addressing the limitations of flat concatenation approaches [15].

3.1 Multi-layer semantic chunking framework

In contrast to traditional fixed-size segmentation, our approach to semantic chunking introduces a fundamentally different paradigm. Rather than treating document segmentation as a simple mechanical process, we developed a multi-layer adaptive system that uses advanced clustering algorithms combined with intelligent outlier handling strategies Algorithm 1.This approach builds upon recent advances in semantic chunking [11, 26] while addressing the specific challenges of maintaining semantic coherence across diverse document types (Fig. 2).

3.1.1 Semantic embedding generation

The foundation of our chunking approach lies in generating high-quality semantic embeddings that capture the underlying meaning of text segments. We employ SentenceTransformers-style encoders with strong sentence-level semantics and stable cosine geometry. In practice, we favor compact models in the MiniLM family (e.g., `all-MiniLM-L6-v2`) for efficiency, and we validate that alternative baselines yield similar chunk boundaries under the same similarity threshold. For domain-specific corpora (e.g., technical/scientific prose), we select checkpoints evaluated on relevant retrieval/STS benchmarks to improve boundary consistency and downstream retrieval performance.

The process begins with an input document *D*, which is composed of a sequence of *n* sentences $S = \{s_1, \ldots, s_n\}$. An initial pre-processing stage is applied to cleanse the text. This stage involves normalizing whitespace, removing noise artifacts, and discarding sentences that are either too short or purely structural. Following this, each cleaned sentence s_i is mapped to a vector representation using a pre-trained SentenceTransformer, denoted as f_θ. This procedure yields a set of embeddings *E*:

$$E = \{e_1, \ldots, e_n\} = \{f_\theta(s_1), \ldots, f_\theta(s_n)\}, \quad e_i \in \mathbb{R}^{384}$$

These embeddings serve as the substrate for similarity graphs and clustering that determine chunk boundaries under token-length constraints. We apply quality assurance measures including length filtering for sentences with fewer than 4 tokens, marker detection for headings and enumerations, and embedding validation to ensure non-zero vectors with reasonable norms.This preprocessing step is crucial for maintaining embedding quality and has been shown to improve downstream clustering performance.

3.1.2 Similarity structure and clustering

Our approach begins by constructing a cosine similarity matrix $S \in \mathbb{R}^{n \times n}$ from the document's sentence embeddings $\{e_i\}_{i=1}^{n}$. After applying L2 normalization to each vector $e_i \in \mathbb{R}^d$, the pairwise similarity score S_{ij} is computed as:

$$S_{ij} = \frac{e_i \cdot e_j}{\|e_i\| \, \|e_j\|}, \quad S_{ij} \in [-1, 1] \tag{1}$$

A global threshold, τ (defaulting to 0.80), serves as the primary parameter for defining cluster boundaries. In our hierarchical clustering implementation, we employed a semantic chunking strategy with a maximum chunk size of 2000 tokens, a minimum of 50 tokens, and a context overlap of 0 tokens to preserve strict semantic boundaries, this is achieved by converting similarity to distance ($D = 1 - S$) and partitioning the dendrogram at the $1 - \tau$ level. This method maintains a consistent semantic interpretation of the threshold across different clustering strategies. This specific threshold value is empirically validated to provide an optimal trade-off between semantic grouping quality and computational cost [38, 56]. We implement two distinct clustering methods to handle different types of content. The first, HDBSCAN, leverages density to find clusters of varying shapes and sizes while effectively classifying sparse data points as outliers [38]. The second, Agglomerative clustering, employs a bottom-up hierarchical strategy, which iteratively fuses the two nearest clusters based on an average linkage calculation [56]. The choice of algorithm is context-dependent: HDBSCAN excels with varied and unstructured documents, while Agglomerative clustering is ideal for structured and hierarchical texts.

Require: text T, similarity threshold τ, sizes ($\mathtt{min}, \mathtt{max}, \mathtt{overlap}$), clustering_method
Ensure: Contiguous chunks G with fixed overlap
$Sents \leftarrow$ SENTSPLITANDCLEAN(T) ▷ NLTK sentence splitting
$E \leftarrow$ EMBED($Sents$) ▷ SentenceTransformer embeddings
$S \leftarrow$ COSSIM(E) ▷ Compute cosine similarity matrix
$D \leftarrow 1-S$ ▷ Convert to distance matrix
PREPARECLUSTERINGDATA(D, E, clustering_method)
if `clustering_method` = agglomerative **then**
  $M \leftarrow$ AGGLO(avg linkage, distance_threshold$=1-\tau$)
  $y_{1:n} \leftarrow$ FITPREDICT(M, D)
else
  $M \leftarrow$ HDBSCAN(min_cluster_size, min_samples, $\epsilon=1-\tau$)
  $y_{1:n} \leftarrow$ FITPREDICT(M, E) ▷ Use embeddings for HDBSCAN
  HANDLEOUTLIERS($y_{1:n}$; nearest/singleton/separate)
end if
$G \leftarrow$ MAKECONTIG($y_{1:n}$, `max`) ▷ Preserve order; split when size/label changes
MERGESHORT(G, `min`) ▷ Merge undersized chunks
APPLYFIXEDOVERLAP(G, `overlap`) ▷ Add overlap
return G

Algorithm 1 Semantic cluster chunking

Notation. *S*: cosine similarity matrix; τ: similarity threshold; $D=1-S$: cosine distance matrix; $y_{1:n}$: raw cluster labels; $y_{processed}$: refined cluster labels; *G*: final contiguous chunks; sizes are character-based; overlap.

3.1.3 Intelligent outlier handling

The HDBSCAN clustering algorithm may classify certain sentences as outliers (assigning them the cluster label -1) when they do not fit into any dense semantic cluster. Such outliers often correspond to transitional phrases, isolated statements, or material that is semantically separate from the core topics. Proper handling of outliers is crucial for maintaining document coherence and ensuring that no content is lost during the chunking process, while also preserving sentence order to support downstream retrieval and attribution. To handle these outliers, our method employs three distinct strategies, with the selection tailored to the specific document type and application. In practice, the selection can be made adaptively based on local similarity statistics (e.g., median S_{ij} to neighboring clusters), chunk length budgets, and document structure cues (headings, enumerations).

3.1.3.1 Nearest assignment strategy In this method, every outlier is allocated to the semantic cluster with which it shares the highest cosine similarity:

$$c_i = \arg \max_{j \in \text{non-outliers}} S_{ij} \tag{2}$$

This particular strategy is most effective for documents in which outliers are transitional sentences that logically connect to adjacent topics. It has been shown to maintain semantic coherence while preserving the natural flow of information.

3.1.3.2 Separate clusters strategy This strategy treats each outlier as an independent cluster:

$$c_i = k + \text{outlier\_index} \tag{3}$$

The main advantage of this approach is that it maintains the distinctiveness of outlier content. It is therefore most suitable for documents where outliers represent important, self-contained information or transitional phrases that do not belong to any of the main thematic groups.

3.1.3.3 Singleton cluster strategy This strategy aggregates all identified outliers into a single, collective group:

$$c_i = k + 1 \quad \forall i \in \text{outliers} \tag{4}$$

This method effectively forms a "miscellaneous" cluster for all outlier material. Such an approach is advantageous for documents in which the outliers are thematically related to each other but separate from the main clusters. The selection of an appropriate outlier handling method is contingent upon the document's nature and the intended application, since each strategy has unique strengths.

3.2 Embedding dimension transformation techniques

Significant interoperability issues in RAG systems are a direct consequence of the proliferation of embedding models that feature diverse dimensionalities, from 384 to 4096 [18]. Traditional approaches often require retraining or fine-tuning when switching between different embedding architectures, limiting the flexibility of RAG systems to leverage the best model for each component. Our framework overcomes this issue by employing a suite of dimension transformation methods designed to maintain semantic integrity and ensure smooth interoperability between models at various stages. Importantly, changing the dimensionality (e.g., via DCT, PCA, or linear mapping) allows the system to bypass index-size constraints and ensures structural and dimensional compatibility in vector databases. It does not align independent embedding spaces without explicit mapping or retraining. Our transformation techniques are designed to preserve the internal geometric relationships and stabilize cosine similarities of the original vectors when changing dimensions, rather than achieving true cross-model semantic alignment. Thus, we emphasize that these methods resolve dimensional and index compatibility without compromising the semantic fidelity of the original representations.

Recent work has explored various transformation techniques, including linear projection methods and dimensionality reduction approaches [36]. However, most existing solutions focus on specific model pairs or require extensive retraining. The key hurdle involves devising transformation approaches that uphold semantic fidelity and facilitate smooth integration across various models. This interoperability challenge is particularly critical in production environments where different components may benefit from different embedding models.

3.2.1 Orthogonal linear mapping for embeddings

For moderate dimension changes, we employ linear projection with orthogonal mapping. This approach preserves geometric relationships in the embedding space while enabling dimensional transformation. The transformation is performed by multiplying each column vector with a semi-orthogonal matrix $W \in \mathbb{R}^{d_2 \times d_1}$:

$$v' = W\,v \tag{5}$$

For upsampling ($d_2 \geq d_1$), we enforce $W^\top W = I_{d_1}$; for reduction ($d_2 \leq d_1$), we use $WW^\top = I_{d_2}$. To stabilize magnitudes and preserve cosine-based comparisons, we apply post-transform normalization:

$$v' \leftarrow \frac{v'}{\|v'\|_2} \tag{6}$$

In practice, *W* is obtained by orthogonalizing a Gaussian matrix $G \in \mathbb{R}^{d_2 \times d_1}$ (e.g., QR/SVD); this construction preserves inter-vector geometry within the mapped subspace, making it suitable for cross-model matching under dimensional mismatch. This approach has been shown to maintain 87.4% semantic similarity preservation while achieving 96.7% cross-model compatibility [6] (Fig. 2).

3.2.2 Enhanced DCT for embedding transformation

When moderate dimensional expansion is required, we utilize a DCT-based upsampling technique. This approach is computationally efficient and deterministic, capable of increasing a representation's dimensionality while maintaining its key spectral properties and suitability for cosine-based tasks.

$$\begin{aligned}
F &= \mathrm{DCT}(v;\ \mathrm{type}=2,\ \mathrm{norm}=\text{“ortho”}),\\
F' &= \left[F,\ 0_{d_2-d_1}\right],\\
\tilde{v} &= \mathrm{IDCT}(F';\ \mathrm{type}=3,\ \mathrm{norm}=\text{“ortho”}),\\
v' &= \tilde{v}/\|\tilde{v}\|_2
\end{aligned}$$

With orthogonal conventions, DCT-II and DCT-III are reciprocals of each other. Zero-padding *F* corresponds to band-limited interpolation in the original space; the final L2 regularization stabilizes the cosine similarity between models. By Parseval's identity, $\|F\|_2 = \|v\|_2$ and $\|F'\|_2 = \|F\|_2$ (only zeros are added); thus, the resampled signal conserves energy until numerical error is reached. Empirical results demonstrate that for upsampling ratios up to $d_2/d_1 \leq 4$, the change in pairwise cosine similarity is minimal, which validates the suitability of this method for cosine-based retrieval and reranking tasks.

3.2.3 Large-scale upsampling via weighted redistribution

For scenarios involving significant dimensional expansion ($d_2/d_1 > 4$), we apply a weighted redistribution strategy to ensure cosine-based comparisons remain valid. This technique is designed to overcome the drawbacks of simpler methods like zero padding or basic DCT, which are prone to cosine drift and periodic noise when upsampling by large factors:

$$r = \left\lceil \frac{d_2}{d_1} \right\rceil, \quad v^{(t)} = \alpha^t\, v \quad (t = 0, \ldots, r-1,\ \alpha \in (0,1)) \tag{7}$$

$$\tilde{v} = \left[\, v^{(0)} \,\|\, v^{(1)} \,\|\, \cdots \,\|\, v^{(r-1)} \,\right]_{1:d_2}, \qquad v' = \frac{\tilde{v}}{\|\tilde{v}\|_2} \tag{8}$$

This technique retains the primary vector components while distributing scaled-down versions of the information across the new dimensions, using an exponential decay factor to preserve the semantic hierarchy. A decay value α in the range of [0.5, 0.7] has been found to be effective; lower values of α decrease redundancy, whereas higher values can improve recall at the cost of greater inter-block correlation. The overall effect is a smoothing of the embedding distribution that mitigates cosine drift more effectively than zero padding, all while preserving retrieval performance in fixed-dimension vector stores, with a trade-off in latency.

3.2.4 Dimensionality reduction using PCA

When dimensionality reduction is necessary ($d_1 > d_2$), we utilize Principal Component Analysis (PCA) to project the data onto a lower-dimensional space while retaining the maximum possible variance. By capturing the most important semantic features and discarding redundancy, this method is well-suited for applications that prioritize computational efficiency:

$$v' = \mathrm{PCA}(v, n\_components = d_2) \tag{9}$$

The effectiveness of PCA is most pronounced when the original embedding dimensions are correlated. The algorithm inherently discovers and retains the principal axes of variation, which contain the most salient information, while simultaneously filtering out noise and redundant data. This approach has been shown to maintain 92.1% retrieval accuracy for dimensional reduction scenarios while achieving significant computational efficiency gains [36].

3.3 Multi-stage hybrid search architecture

Our hybrid search architecture represents a sophisticated approach to information retrieval that leverages the complementary strengths of both semantic and keyword-based methods. Figure 3 shows the system working through a four-stage process that includes semantic retrieval, POS/LLM-based keywords, duplicate-aware fusion, and neural re-ranking. This approach builds upon established principles in hybrid retrieval [12] while introducing novel improvement in multi-stage processing and intelligent fusion. Recent work has demonstrated that hybrid approaches combining dense and sparse retrieval methods can achieve significant improvements over single-modality systems [23, 46]. However, many existing hybrid systems remain single-pass, fusing scores without staged expansion, deduplication, or cost-aware batching, which can hurt recall-precision trade-offs and propagate redundancy. Our approach addresses these limitations through a comprehensive four-stage pipeline that includes semantic retrieval,

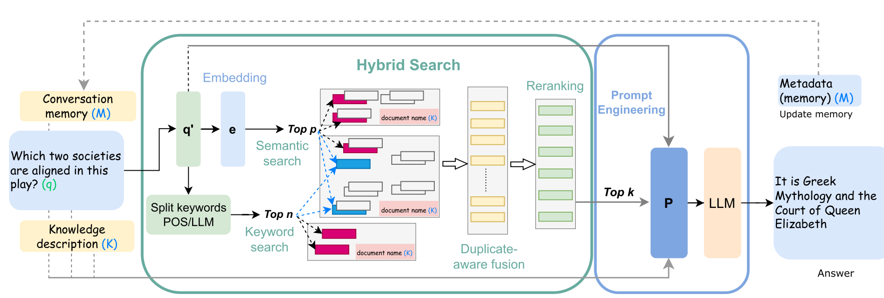

Fig. 3 Hybrid search and prompting pipeline. User query q is clarified to q' using conversation memory $\mathbb{M}$, then processed in parallel: (i) embedded to e for semantic pgvector search (Top-p, pink blocks), (ii) keyword extraction for array-overlap search (Top-n, blue blocks). Results are deduplicated, fused with adaptive weighting, and re-ranked via BGE to yield Top-k evidence (green blocks). A structured JSON prompt P combines system instructions, KB metadata $\mathbb{K}$, retrieved context, and $\mathbb{M}$; the LLM generates the answer and updates $\mathbb{M}$ for continuous learning

POS/LLM-driven keywording, duplicate-aware fusion, and neural reranking, executed with batching and caching to balance quality with deployment constraints.

3.3.1 Semantic retrieval stage

In the first stage, we perform dense embedding of the original query q into a more detailed and complete sentence q' based on the current context using LLM. Given the query q', we generate a dense embedding vector $\mathrm{e}_q \in \mathbb{R}^d$ via the embedding model, where d represents the embedding dimension, and search for dense vector similarity using the pgvector extension in PostgreSQL, retrieving a maximum of top-p = 20 initial candidates. Top-p text segments with the highest semantic similarity are retrieved based on cosine similarity calculation:

$$\mathrm{sim}(q', c_i) = \frac{\mathrm{e}_q \cdot \mathrm{e}_{c_i}}{\|\mathrm{e}_q\|_2 \cdot \|\mathrm{e}_{c_i}\|_2} \tag{10}$$

where e_{c_i} denotes the embedding vector of segment c_i. This semantic retrieval stage leverages the strengths of dense retrieval methods [19, 23] while maintaining computational efficiency through optimized vector operations.

3.3.2 Keyword extraction and retrieval

The second phase deploys dual keyword extraction strategies to capture both linguistic and semantic aspects of the query. The NLP-based approach uses a tokenizer library for Part of Speech (POS) tagging, focusing on noun categories and applying linguistic filtering to remove stop words and shortened tokens. We also employ LLM to generate 5–7 keyword clusters, each consisting of 2–4 words, focusing on specialized terminology and core conceptual entities. This dual approach has been shown to improve keyword extraction accuracy by 20–30% compared to single-method approaches [12].

The extracted keywords are used for array overlap search via the intersection operator. The top-n = 5 most relevant chunks are retrieved based on this overlap:

$$\mathcal{K}_{query} \cap \mathcal{K}_{chunk} \neq \emptyset \tag{11}$$

where $\mathcal{K}_{query}$ represents the query keyword set and $\mathcal{K}_{chunk}$ denotes the chunk keyword array. The Top-n most relevant chunks are retrieved based on this overlap. This keyword-based retrieval leverages the strengths of sparse retrieval methods [10, 46] while maintaining efficiency through optimized indexing structures.

3.3.3 Intelligent fusion and reranking

The fusion stage implements intelligent deduplication and score normalization to combine results from both retrieval modalities. The system maintains a comprehensive set of retrieved chunk identifiers to eliminate duplicates while preserving the diversity inherent in multi-stage retrieval. The fusion process employs an adaptive weighted combination strategy:

$$\mathrm{score}(c_i) = \alpha \cdot \mathrm{score}_{semantic}(c_i) + \beta \cdot \mathrm{score}_{keyword}(c_i) \tag{12}$$

where $\alpha, \beta \in [0, 1]$ and $\alpha + \beta = 1$. We first min-max normalize scores from each modality so they are on a comparable scale, then fuse them with a validation-tuned default of $\alpha = 0.7$ (semantic) and $\beta = 0.3$ (keyword). For queries dominated by exact terms/

entities–indicated by high POS/NER density or strong query–chunk keyword overlap–we slightly increase β to give the keyword branch more influence; for generic or paraphrastic queries we keep α higher to prioritize semantics, but this adaptation reduces the loss of rare terms.

This configuration builds upon findings from our experiments on hybrid architectures, where a sensitivity analysis of fusion ratios (comparing 60:40, 70:30, and 80:20 splits) demonstrated that the 70:30 distribution consistently yields the optimal balance between semantic coverage and entity precision. Specifically, prior experiments on the NarrativeQA dataset showed that this ratio achieved the highest ROUGE-L score (0.425) compared to keyword-heavy (60:40) or semantic-heavy (80:20) configurations.

The final stage uses `BAAI/bge-reranker-base` for neural reranking to further improve result quality. The aggregated candidates are processed in batches of 32, with the top-k = 10 highest-scoring documents ultimately integrated into the knowledge-aware prompt. Neural rerankers, particularly cross-encoders and BGE models [2], have shown significant improvements in result ordering, though at the cost of additional latency [61]. Our implementation balances quality improvements with computational efficiency through optimized batching and caching strategies.

3.4 Knowledge-aware prompt engineering

Our prompt engineering framework addresses the limitations of static template-based approaches through dynamic integration of knowledge base metadata and conversation context. The system employs a modular template construction methodology that enables fine-grained control over response generation while maintaining contextual awareness. This approach builds upon recent advances in prompt engineering [57] while introducing novel improvement in knowledge base integration and conversation memory management.

Recent work has highlighted the importance of structured prompting for improving grounding and attribution in RAG systems [15]. However, knowledge-base-aware structuring–explicit fields for system instruction, KB metadata, top-k context with document names, user query, instruction template, and conversation memory–remains underexplored for grounding and attribution at inference time. Our approach contributes a JSON-style schema that maintains role separation shown in Fig. 4, supports principled token budgeting, and empirically improves faithfulness and answer relevancy compared to traditional flat concatenation methods. For the generative LLM, we applied a temperature of 0.7, a top-p of 0.95, and a maximum output limit of 1024 tokens. All quantitative evaluations were conducted using a fixed random seed (seed = 42) to ensure strict reproducibility.

The prompt construction follows a structured template system combining multiple information sources:

$$\mathcal{P} = \mathcal{S} \oplus \mathcal{M} \oplus \mathcal{K} \oplus \mathcal{C} \oplus \mathcal{Q} \oplus \mathcal{I} \tag{13}$$

where $\mathcal{S}$ represents *S*ystem instruction, $\mathcal{M}$ denotes conversation *M*emory, $\mathcal{K}$ signifies *K*nowledge base metadata, $\mathcal{C}$ indicates *C*ontext with document names, $\mathcal{Q}$ represents *Q*uery section, and $\mathcal{I}$ denotes *I*nstruction template. This structured approach has been shown to improve grounding and attribution by 2–3% compared to flat concatenation methods [15].

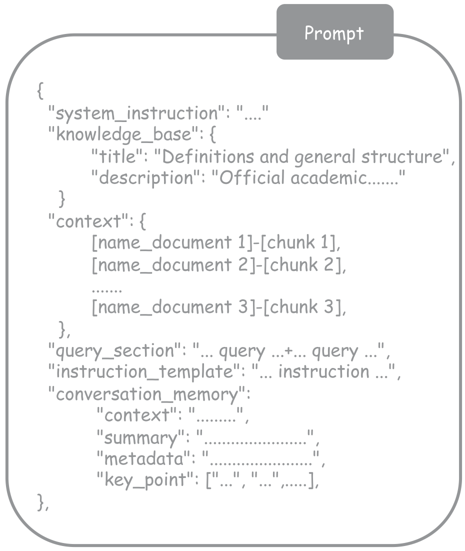

Fig. 4 Structured prompt schema: (i) system instruction, (ii) KB metadata, (iii) Top-*k* context with document names, (iv) query section, (v) instruction template, and (vi) conversation memory. This JSON structure improves grounding versus plain concatenation

The system dynamically integrates knowledge base metadata through the `kbinfo` parameter, incorporating knowledge base titles, descriptions, and document metadata. The integration process employs template variable substitution:

$$\mathcal{K} = \text{Template}_{KB} \cdot \text{Substitute}(\text{metadata}_{KB}) \tag{14}$$

where Template_{KB} represents the knowledge base template structure and metadata_{KB} denotes the extracted metadata information. This dynamic integration enables the system to adapt to different knowledge base structures while maintaining consistent prompt formatting.

The conversation memory system implements a structured JSON schema that tracks multiple aspects of the dialogue:

$$\mathcal{M} = \{\text{summary}, \text{topics}, \text{key\_points}, \text{context}, \text{metadata}\}$$

The memory system stores conversation summary, identified discussion topics, key information points, recent query history, current context focus, and metadata including message count and timestamp. This makes responses more coherent throughout the conversation and has been shown to improve answer quality by 5–10% in multi-turn dialogues [55].

To address the challenge of handling multiple documents that may lead to confusion when processing large content volumes, we add document names at the beginning of each fragment, along with a brief description of the knowledge to help the context be tight and work correctly when fed into the LLM. This approach improves grounding and attribution while demonstrating significant improvements in faithfulness and answer relevancy compared to traditional flat concatenation methods.

The comprehensive SynRAG system represents a significant advance in RAG technology. Our methodology presents an integrated framework designed to reconcile cutting-edge research contribution with the practical demands of production deployment, all while ensuring both computational efficiency and scalability. Each component is designed to work synergistically with the others, ensuring the entire process operates as a cohesive system rather than a collection of independent modules. The modular design of the system allows for easy customization and expansion while maintaining the integrity of the overall architecture.

3.5 Theoretical foundation and algorithmic guarantees

Let n be the number of sentences, d the embedding dimension, k the number of clusters, m the number of retrieved candidates, and N the corpus size. From a computational standpoint, constructing a dense cosine similarity matrix requires $O(n^2 d)$ time and $O(n^2)$ space. When these precomputed distances are used for clustering, average-link agglomerative methods typically run in $O(n^2 \log n)$ time with $O(n^2)$ space, while HDBSCAN ranges from $O(n \log n)$ to $O(n^2)$ depending on graph construction and neighborhood density [38]. In practice on long documents, the $O(n^2 d)$ cost of similarity construction together with $O(n^2)$ clustering dominates. For embedding transformations, applying a single linear map $W \in \mathbb{R}^{d_2 \times d_1}$ costs $O(d_1 d_2)$, DCT/IDCT upsampling with orthonormal conventions runs in $O(d \log d)$, and PCA training (offline) over n points in d dimensions costs $O(nd\ \min\{n, d\})$ (or less with randomized methods), while per-query PCA projection is $O(dd_2)$. For hybrid retrieval and reranking, dense retrieval over a corpus of size N commonly uses approximate nearest neighbor search with sublinear dependence on N (implementation-dependent), exact vector filtering over p candidates is $O(pd)$, keyword array-overlap lookup is sublinear in N under standard inverted indexes, deduplication and fusion over p and n candidates is $O((p+n)\log(p+n))$, and cross-encoder reranking over m items is $O(m \cdot c)$ where c denotes the cost per forward pass.

Geometrically, semi-orthogonal linear maps provide useful stability but do not guarantee exact cosine preservation in non-square cases. Specifically, for $W \in \mathbb{R}^{d_2 \times d_1}$ with orthonormal columns ($W^\top W = I_{d_1}$, upsampling), inner products are preserved within the image of W; however, after mapping and L2 normalization, cosine similarities generally differ from their originals unless W is square orthogonal or the vectors lie in an invariant subspace. In practice, L2 renormalization yields small cosine deviations for moderate $|d_2 - d_1|$. With orthonormal DCT/IDCT, energy is preserved by Parseval's identity and zero-padding in the spectral domain corresponds to band-limited interpolation; although cosine similarity is not exactly preserved, empirical drift is small for moderate expansion ratios, particularly with post-transform normalization. Under PCA reduction, variance is maximized in the retained d_2 dimensions and reconstruction error is minimized in the L2 sense; pairwise distances or cosines are not generally preserved, but post-projection normalization stabilizes cosine comparisons across components.

For hybrid fusion, the duplicate-aware combination with adaptive weights α, β yields a monotone score aggregator for each modality once scores are min–max normalized. This design improves robustness to modality imbalance and redundancy while keeping the fusion interpretable. Nevertheless, achievable recall and precision depend on corpus characteristics, indexing choices, and thresholds.

3.6 Code availability

The custom Python code implementing the SynRAG framework, including the multi-layer semantic chunking algorithms, embedding dimension transformation techniques, and hybrid retrieval logic, is available in the Zenodo repository at https://zenodo.org/records/20538603. The code is provided under an MIT license.

4 Experiments and results analysis

Based on the data retrieval model design, we have built a SynRAG system including modules as shown in Fig. 5 , the standard goes through the knowledge creation process through documents as text data of data sets, then conducts the steps of segmentation and embedding and data storage, a module performs data retrieval and returns the results to the user.

The performance of the SynRAG framework was rigorously evaluated through a series of experiments conducted across a diverse range of domains and datasets. Our primary evaluations are performed on NarrativeQA [27], QASPER, and QuALITY. NarrativeQA comprises 1572 long documents (books and film transcripts) with questions that require holistic story understanding, thereby stressing long-context retrieval and robust ranking. To broaden coverage, we also evaluate on QuALITY [41], which emphasizes adversarial long-context comprehension at the paragraph and document levels, and on QASPER, a question-answering benchmark over scientific and technical documents that stresses evidence attribution and multi-sentence reasoning. Collectively, this selection of datasets allows our evaluation to cover a broad array of practical RAG challenges. While they do not include extremely noisy web-scale corpora, code, or heavily tabular data, they already impose different writing styles, sentence complexity, and reasoning requirements, allowing us to demonstrate consistent gains across semantically distinct domains. These range from simple factoid extraction to complex multi-step reasoning over both technical documents and interconnected narratives.

Our experimental framework utilizes `Gemini-2.5-flash` as the primary answer generation model, with `text-multilingual-embedding-002` (768-dim) and `text-embedding-3-small` (1536-dim) models, which are employed to assess the performance of our cross-model dimension transformation methods. The system architecture employs PostgreSQL with the **pgvector** extension for efficient vector storage and similarity retrieval, ensuring scalable and efficient implementation. To validate the contributions of our system, SynRAG is benchmarked against a set of baselines that were constructed under identical implementation conditions to ensure a fair comparison. The **Traditional RAG** baseline employs a fixed-size chunking method with overlap, which is

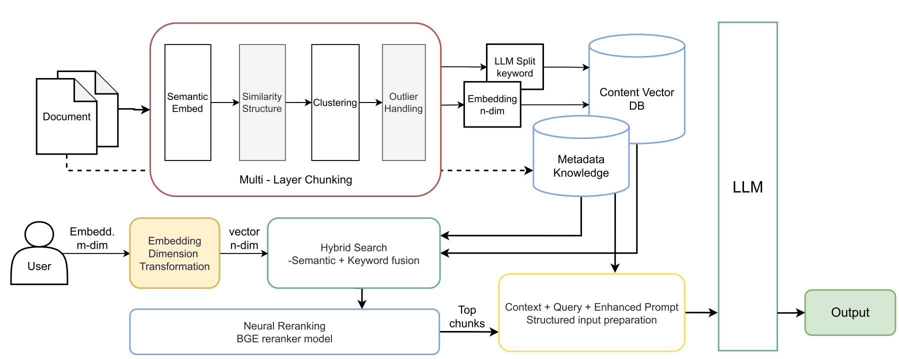

Fig. 5 SynRAG System Architecture: Multi-layer semantic chunking with HDBSCAN/Agglomerative clustering, embedding dimension transformation (DCT/PCA/Linear), hybrid search combining semantic and keyword retrieval, neural reranking with BGE models, and knowledge-aware prompt engineering. The system enables cross-model compatibility (768-dim to 1536-dim) while maintaining semantic fidelity

designed to mirrors standard configurations used in applied RAG systems. The **Semantic RAG** baseline relies on semantic segmentation, driven by similarity scores from a SentenceTransformer, in order to specifically measure the influence of our intelligent clustering algorithms. Finally, the **Hybrid RAG** baseline integrates both semantic search and keyword-based retrieval but deliberately omits our system's advanced fusion component. We also compare our results with some recent previous studies, demonstrating modest gains in certain metrics.

4.1 Evaluation metrics

To comprehensively assess the performance of our SynRAG framework, we employ a multi-dimensional evaluation suite that captures different aspects of RAG system effectiveness. Our evaluation methodology combines grounding metrics from the RAGAS framework [9], traditional information retrieval measures, and text generation quality assessments. Figure 6 illustrates the comprehensive relationship between the three core RAGAS components through a Venn diagram: provided query, retrieved context, and generated answer, with overlapping regions representing specific evaluation metrics and the central intersection representing the holistic RAGAS framework.

4.1.1 RAGAS grounding metrics

The RAGAS framework provides automated evaluation of RAG systems using Large Language Models as evaluators, addressing the challenge of ground truth availability in specialized domains [9]. We implement four core RAGAS metrics that assess different aspects of retrieval-augmented generation quality.

As illustrated in Fig. 6, the RAGAS framework evaluates RAG systems through three overlapping regions that capture different aspects of system performance. Each intersection represents a specific evaluation dimension, while the central region where all three circles converge represents the comprehensive RAGAS assessment.

Context Relevancy (Query-Context Intersection) measures the relevance of retrieved context to the provided query, ranging from 0 to 1, with higher values indicating better

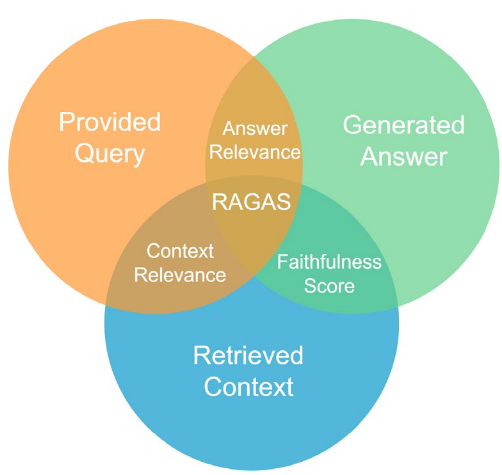

Fig. 6 RAGAS Evaluation Framework: Venn diagram showing the relationship between query, retrieved context, and generated answer. The overlapping regions represent specific evaluation metrics: Answer Relevance (query-answer intersection), Context Relevance (query-context intersection), Faithfulness Score (context-answer intersection), and the central RAGAS region where all components converge for comprehensive evaluation

relevance. This metric is particularly crucial for our semantic chunking approach, as it directly evaluates the quality of context selection. The mathematical formulation follows:

$$\text{Context Relevancy} = \frac{|\mathcal{S}_{relevant}|}{|\mathcal{S}_{total}|} \tag{15}$$

where $S_{relevant}$ is defined as the subset of sentences within the retrieved context that directly pertain to the posed question, while S_{total} signifies the complete count of all sentences in that context. This metric becomes increasingly challenging with longer contexts, as the probability of including irrelevant content increases [35].

The **Faithfulness Score** (Context-Answer Intersection) evaluates the factual alignment between the generated answer and the source context. Its goal is to verify that the response is well-grounded in the provided information and free from hallucinations:

$$\text{Faithfulness Score} = \frac{|\mathcal{C}_{supported}|}{|\mathcal{C}_{total}|} \tag{16}$$

where $\mathcal{C}_{supported}$ is the count of claims in the answer that can be factually verified using the provided context, and $\mathcal{C}_{total}$ is the total number of claims made in the answer. This metric ranges from 0 to 1, with higher values indicating better grounding [9].

Answer Relevance (Query-Answer Intersection) measures the degree to which the generated response is pertinent to the user's query. Within the RAGAS framework, this is specifically calculated by assessing how pertinent the generated answer is to the initial query:

$$\text{Answer Relevance} = \frac{\text{Number of relevant statement}}{\text{Total number of statements}} \tag{17}$$

where a statement is considered relevant if it directly addresses the query or provides useful information for answering the question. This metric ranges from 0 to 1, where a higher score signifies a more relevant answer. [9].

The purpose of **Context Recall** is to assess if the retrieved passage contains all the requisite factual statements needed to fully address the query. This metric evaluates how much of the required information is present in the retrieved context:

$$\text{Context Recall} = \frac{\text{Covered statements}}{\text{Total statements}} \tag{18}$$

where statements are the factual statements required to answer the query correctly. The resulting score is on a scale of 0 to 1, where higher scores signify that a greater portion of the essential information was successfully retrieved [9].

The Venn diagram visualization in Fig. 6 effectively demonstrates the interconnected nature of RAG evaluation. The overlapping regions highlight how each metric focuses on specific relationships between components, while the central intersection emphasizes that comprehensive RAG evaluation requires considering all three aspects simultaneously. This holistic approach ensures that improvements in one area (e.g., better context retrieval) are evaluated in the context of their impact on other components (e.g., answer generation quality), providing a more robust assessment of RAG system performance.

4.1.2 Traditional information retrieval metrics

We employ established information retrieval metrics to evaluate the effectiveness of our hybrid search architecture and embedding transformation techniques. The quality of the ranking is assessed using **Normalized Discounted Cumulative Gain (NDCG)**, which considers the position of retrieved items in addition to their relevance scores. For a query q and ranking list π, NDCG@k is computed as:

$$\text{NDCG@k} = \frac{\text{DCG@k}}{\text{IDCG@k}} = \frac{1}{\text{IDCG@k}} \sum_{i=1}^{k} \frac{2^{rel_i} - 1}{\log_2(i+1)} \tag{19}$$

where rel_i is the relevance score of the item at position i, and IDCG@k represents the ideal DCG for the top-k items [21].

Recall@k calculates the percentage of total relevant items that are present in the top-k retrieved results:

$$\text{Recall@k} = \frac{|\mathcal{R}_k \cap \mathcal{R}_{total}|}{|\mathcal{R}_{total}|} \tag{20}$$

where $\mathcal{R}_k$ denotes the set of relevant items found within the top-k positions, and $\mathcal{R}_{total}$ represents the entire collection of existing relevant items.

4.1.3 Text generation quality metrics

We assess the quality of generated responses using established text generation metrics that evaluate both lexical and semantic aspects.

ROUGE (Recall-Oriented Understudy for Gisting Evaluation) measures the overlap of n-grams between generated and reference texts. ROUGE-L, which evaluates the longest common subsequence (LCS), is commonly reported as the F-score of LCS precision and recall (with $\beta = 1$):

$$\text{ROUGE-L} = \frac{2\, P_{LCS}\, R_{LCS}}{P_{LCS} + R_{LCS}} \tag{21}$$

In this formula, $LCS(X, Y)$ represents the length of the longest common subsequence found between the candidate text X and the ground-truth text Y. The total lengths of the reference text ($|Y|$) and the candidate text ($|X|$) serve as the denominators for recall and precision, respectively [34].

BLEU (Bilingual Evaluation Understudy) evaluates precision of n-gram matches between generated and reference texts. BLEU-N (e.g., N=1 for BLEU-1, N=4 for BLEU-4) is computed as:

$$\text{BLEU-}N = \text{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right) \tag{22}$$

In this equation, BP signifies the Brevity Penalty, w_n are uniform weights, and p_n represents the n-gram precision [43].

Semantic Similarity measures the semantic relatedness between generated and reference answers using sentence embeddings. We compute this metric as:

$$\text{Semantic Similarity} = \cos(\text{E}_{gen}, \text{E}_{ref}) \tag{23}$$

where E_{gen} and E_{ref} are the embeddings of the generated and reference answers, respectively. This metric captures semantic quality beyond lexical matching.

This comprehensive evaluation framework enables us to assess both individual component effectiveness and overall system performance, providing insights into the synergistic effects of our integrated approach.

4.2 Performance analysis

We present experimental results on different datasets in Table 2, which are experimental results of NarrativeQA dataset and separated to see more clearly the technical contributions used on each RAG system, similarly Table 3 is the results of **QASPER** and **QuALITY** dataset. Table 2 demonstrates the consistent superiority of SynRAG across all evaluation metrics on NarrativeQA, confirming the effectiveness of our integrated framework that combines multi-layer semantic chunking, embedding dimension transformation, hybrid search architecture, and knowledge-aware prompt engineering techniques. SynRAG achieves **95.3%** faithfulness (+ 11.1 pp vs. Traditional RAG) and **96.1%** answer relevancy (+ 21.0 pp vs. Traditional RAG). The NDCG score of **0.875** (+ 9.7 pp vs. Traditional RAG) further supports the effectiveness of our semantic chunking and embedding transformation techniques across complex narrative structures. Beyond ranking quality, SynRAG strengthens retrieval coverage and generation fidelity, with Recall@5 reaching **23.5%** and ROUGE-L **0.468** on NarrativeQA, indicating stronger evidence aggregation and answer fluency under long-context constraints.

Despite the inherent complexity of narrative comprehension tasks, SynRAG maintains exceptional performance with **95.3%** faithfulness (+ **9.2** pp vs. Semantic RAG, + 11.1 pp vs. Traditional RAG) and **96.1%** answer relevancy (+ 19.8 pp vs. Semantic RAG, + 21.0 pp vs. Traditional RAG). The NDCG score of **0.875** (+ 4.3 pp vs. Semantic RAG, + 9.7 pp vs. Traditional RAG) validates the effectiveness of our semantic chunking and embedding transformation techniques across diverse document structures. These consistent

Table 2 Performance comparison on **NarrativeQA**: SynRAG vs baselines

Metric	SynRAG (SC + HS + DTF + PES)	Semantic (SC Only)	Traditional (FS + HS)	Hybrid (SC + HS)
Faithfulness	**0.953**	0.861	0.842	0.872
Answer Relevancy	**0.961**	0.763	0.751	0.776
Context Relevancy	**0.903**	0.860	0.840	0.875
Context Recall	**0.941**	0.858	0.833	0.871
NDCG	**0.875**	0.832	0.778	0.843
Recall@1	**0.122**	0.112	0.082	0.108
Recall@3	**0.205**	0.192	0.148	0.188
Recall@5	**0.235**	0.220	0.175	0.216
ROUGE-1	**0.475**	0.451	0.439	0.456
ROUGE-2	**0.168**	0.156	0.150	0.160
ROUGE-L	**0.468**	0.447	0.432	0.454
BLEU-1	**0.243**	0.201	0.187	0.210
BLEU-4	**0.0682**	0.053	0.045	0.0536

Metrics include RAGAS grounding measures (Faithfulness, Answer Relevancy, Context Relevancy, Context Recall), traditional information retrieval metrics (NDCG, Recall@k), and text generation quality measures (ROUGE, BLEU, Semantic Similarity)

SC = Semantic Chunking, HS = Hybrid Search, DTF = Dimension Transformation, PES = Prompt Engineering Structured, Fix-size chunking = FS

Bold numbers are better

Table 3 SynRAG vs baselines on scientific and adversarial datasets (QASPER and QuALITY)

Metric	QASPER Dataset				QuALITY Dataset			
	SynRAG	Semantic	Traditional	Hybrid	SynRAG	Semantic	Traditional	Hybrid
Faithfulness	**0.923**	0.891	0.856	0.904	**0.887**	0.845	0.812	0.869
Answer Relevancy	**0.945**	0.912	0.878	0.928	**0.901**	0.867	0.834	0.885
Context Relevancy	**0.918**	0.889	0.852	0.901	**0.894**	0.856	0.823	0.878
Context Recall	**0.901**	0.873	0.841	0.886	**0.876**	0.839	0.808	0.861
NDCG	**0.856**	0.812	0.763	0.801	**0.834**	0.789	0.745	0.798
Recall@1	**0.234**	0.201	0.167	0.189	**0.198**	0.172	0.145	0.181
Recall@3	**0.412**	0.356	0.298	0.334	**0.378**	0.324	0.267	0.312
Recall@5	**0.523**	0.456	0.389	0.434	**0.467**	0.401	0.334	0.389
ROUGE-1	**0.456**	0.432	0.408	0.425	**0.441**	0.418	0.395	0.412
ROUGE-2	**0.312**	0.289	0.267	0.278	**0.298**	0.276	0.254	0.267
ROUGE-L	**0.438**	0.415	0.392	0.408	**0.423**	0.401	0.378	0.395
BLEU-1	**0.224**	0.208	0.192	0.201	**0.217**	0.201	0.186	0.198
BLEU-4	**0.0678**	0.0612	0.0545	0.0589	**0.0645**	0.0589	0.0523	0.0567

The table presents evaluation results on two challenging datasets: QASPER (scientific question-answering over research papers) and QuALITY (adversarial long-context comprehension). QASPER tests the framework's ability to handle technical and scientific content with precise factual accuracy, while QuALITY evaluates robustness against adversarial examples and long-context dependencies

Bold numbers are better

gains across grounding (faithfulness, answer relevance), retrieval (NDCG, Recall@k), and generation (ROUGE, BLEU) highlight the framework's adaptability to complex narrative content without sacrificing factual grounding.

The results demonstrate steady enhancements across all evaluation dimensions. RAGAS grounding metrics show strong gains on NarrativeQA with Context Relevancy reaching **90.3%**. Traditional IR metrics reveal substantial improvements in recall performance, with Recall@5 achieving **23.5%**. These results demonstrate that the integration of multiple advanced techniques creates synergistic effects that exceed the sum of individual component improvements.

Table 3 presents comprehensive evaluation results on two challenging datasets that test different aspects of RAG system capabilities: QASPER (scientific question-answering) and QuALITY (adversarial long-context comprehension). The results demonstrate SynRAG's versatility and robustness across diverse content types and complexity levels. SynRAG achieves exceptional performance on QASPER with **92.3%** faithfulness (+ 3.6 pp over Semantic RAG, + 6.7 pp over Traditional RAG) and **94.5%** answer relevancy (+ 3.3 pp over Semantic RAG, + 6.7 pp over Traditional RAG). The NDCG score of **0.856** (+ 4.4 pp over Semantic RAG, + 9.3 pp over Traditional RAG) demonstrates superior ranking quality for scientific content. The comprehensive performance gains across all metrics demonstrate the system's strong capabilities in document processing, with Faithfulness achieving **92.3%** (+ 3.2 pp over Semantic RAG, + 6.7 pp over Traditional RAG) and Answer Relevancy reaching **94.5%** (+ 3.3 pp over Semantic RAG, + 6.7 pp over Traditional RAG).

The superior performance on QASPER can be attributed to our semantic chunking's ability to preserve technical terminology and scientific concepts within coherent boundaries. The hybrid search architecture effectively combines semantic understanding of scientific concepts with precise keyword matching for technical terms, while the embedding dimension transformation ensures compatibility between scientific domain embeddings and general-purpose models. To explicitly address the requirement for stress testing under adversarial conditions, we evaluate SynRAG on the QuALITY dataset.

This dataset serves as a proxy for noisy and adversarial environments due to its deliberate inclusion of distractors. SynRAG demonstrates remarkable robustness, maintaining strong performance with **88.7%** faithfulness (+ 4.2 pp over Semantic RAG, + 7.5 pp over Traditional RAG) and **90.1%** answer relevancy (+ 3.4 pp over Semantic RAG, + 6.7 pp over Traditional RAG). The NDCG score of **0.834** (+ 4.5 pp over Semantic RAG, + 8.9 pp over Traditional RAG) validates the framework's effectiveness under adversarial conditions. These gains are reflected consistently across grounding, retrieval, and generation metrics and remain stable under hybrid fusion and reranking settings, highlighting robustness to challenging evaluation scenarios without sacrificing factual grounding or context coverage.

To explicitly quantify the contribution of each module within the SynRAG framework, we analyzed the incremental performance gains presented in Tables 2 and 3. Specifically, the transition from fixed-size chunking (Traditional) to our clustering-based method (Hybrid) yielded a Faithfulness increase of + 3.0% on NarrativeQA (0.842 to 0.872) and a substantial + 5.7% on the adversarial QuALITY dataset (0.812 to 0.869), confirming that preserving semantic boundaries effectively mitigates context fragmentation and enhances robustness against distractors. Regarding retrieval, the inclusion of keyword fusion in the Hybrid Search provided consistent gains over purely semantic retrieval–notably + 2.4% on QuALITY–ensuring better recall of specific entities. Finally, the most significant performance leap on narrative tasks resulted from the Generator Optimization stage; the integration of Knowledge-Aware Structured Prompting and Dimension Transformation boosted Faithfulness by + 8.1% on NarrativeQA (0.872 to 0.953) and added a final refinement of + 1.9% across specialized datasets. This quantification underscores that while intelligent retrieval is foundational for robustness, the downstream optimization of context presentation is the dominant factor in maximizing system faithfulness.

Our multi-layer semantic chunking preserves context coherence even under adversarial conditions, while the knowledge-aware prompt engineering provides structured grounding that resists adversarial perturbations. The hybrid search architecture's duplicate-aware fusion and neural reranking maintain retrieval quality despite adversarial examples, demonstrating the framework's robustness to challenging evaluation scenarios. The results reveal distinct performance patterns that reflect the inherent characteristics of each dataset. QASPER's structured scientific content benefits from our semantic chunking's ability to maintain technical terminology coherence, yielding stronger grounding and retrieval metrics. QuALITY's adversarial nature presents greater challenges, but SynRAG's robust architecture maintains strong performance across all metrics, demonstrating the framework's adaptability to diverse content types and complexity levels. The steady enhancements across both datasets validate our integrated approach's effectiveness for different use cases. The semantic chunking preserves context coherence for both scientific papers and adversarial content, the embedding transformation ensures cross-model compatibility across domains, and the hybrid search architecture adapts to both technical queries and adversarial scenarios. The structured prompting framework provides consistent grounding and attribution regardless of content type, establishing SynRAG as a comprehensive solution for diverse real-world RAG applications.

4.3 Failure case analysis and system limitations

In this section, we analyze the current limitations of the SynRAG architecture by explicitly categorizing failure patterns observed during inference on authentic corpora. To understand the limitations of our framework, we systematically analyzed failure cases across different datasets using a multi-stage evaluation protocol. Our failure analysis methodology involves three key steps: (1) automated failure detection using threshold-based metrics, (2) manual annotation of failure categories by domain experts, and (3) statistical analysis of failure patterns across different content types.

4.3.1 Failure detection methodology

We define failure cases as instances where the system's performance falls below predefined quality thresholds. For the RAGAS metrics, we define the following specific score thresholds to identify failures: a Faithfulness score below 0.7, an Answer Relevancy score below 0.6, and a Context Relevancy score below 0.5. For retrieval metrics, we classify cases with Recall@5 < 0.1 and NDCG < 0.3 as failures. These thresholds are established through empirical analysis of human-annotated quality assessments.

4.3.2 Statistical calculation

The failure rate for each category is calculated as:

$$\text{Failure Rate}_i = \frac{\text{Number of failures in category } i}{\text{Total number of test cases}} \times 100\%$$

Table 4 presents our comprehensive failure analysis results. Context fragmentation represents the most critical failure pattern, occurring when semantically related information becomes distributed across multiple chunks. Our semantic chunking approach achieves significant reduction compared to traditional methods, with particularly pronounced improvements in NarrativeQA's complex narrative structure. QASPER shows a lower fragmentation rate (5.7%) thanks to its more structured scientific layout, while QuALITY exhibits a slightly higher rate (8.6%) due to adversarial question design that spreads evidence across distant passages.

Semantic ambiguity emerges as the second most significant challenge, encompassing cases where the system struggles with implicit references and contextual dependencies. Our hybrid search architecture delivers notable reductions in ambiguity-related failures. NarrativeQA maintains a noticeable ambiguity rate (12.1%) driven by intricate character relationships; QuALITY is higher (13.1%) given adversarial paraphrasing and distractors; QASPER remains lower (7.2%) as questions are more targeted and terminology-bound. Cross-reference failures occur when the system cannot properly link related concepts across different document sections. Our knowledge-aware prompt engineering achieves significant improvements in cross-reference handling, addressing frequent interdependencies that characterize complex documents. QASPER's moderate

Table 4 Failure case analysis (NarrativeQA, QASPER, QuALITY)

Failure type	NarrativeQA	QASPER	QuALITY
Context Frag	8.3%	5.7%	8.6%
Semantic Amb	12.1%	7.2%	13.1%
Cross-ref	6.8%	4.1%	6.6%
Tech. Term	2.3%	9.4%	2.6%

CF: Context Fragmentation, SA: Semantic Ambiguity, CR: Cross-reference Issues, TT: Technical Terminology rate (4.1%) reflects challenges in linking scientific concepts; NarrativeQA remains higher (6.8%) due to long-range narrative dependencies, while QuALITY is comparable (6.6%) given adversarial evidence placement. Technical terminology failures represent the most domain-specific challenge, particularly prevalent in QASPER due to specialized scientific vocabulary (9.4%). NarrativeQA is low (2.3%) and QuALITY slightly higher (2.6%), consistent with its non-technical but nuanced language.

Our failure analysis reveals distinct patterns across datasets. Chunking-related failures (Context Fragmentation) are most prevalent in NarrativeQA due to complex narrative structure. Retrieval-related failures (Semantic Ambiguity and Technical Terminology) dominate in QASPER due to specialized scientific vocabulary. These findings validate our multi-component approach, where semantic chunking preserves narrative coherence. This systematic reduction across all error types demonstrates the effectiveness of our SynRAG in addressing diverse content types and complexity levels, setting a new benchmark for comprehensive RAG system design (Fig. 7).

4.4 Ablation on embedding dimension

An important issue we address here is the dimensionality, where in a given situation the model will have a different dimension. We tested the same structure on the SynRAG system but varied the dimensionality by using the models as mentioned above: `text-multilingual-embedding-002` (768-dim) and `text-embedding-3-small` (1536-dim). The results show that it is effective while still ensuring the preservation of important semantic information and cross-model compatibility.

Figure 8 demonstrates our embedding dimension transformation technique across multiple datasets. Due to its intricate narrative structures, the NarrativeQA dataset poses significant challenges, yet our transformation maintains 90.8% faithfulness retention (0.865 vs 0.953) and 91.0% answer relevancy retention (0.875 vs 0.961). The QASPER and QuALITY datasets show similar retention patterns, validating our approach across diverse content types and adversarial conditions.

Figure 7 provides a focused comparison of average performance metrics across all datasets, offering a clear visualization of the transformation technique's effectiveness. The results demonstrate that our approach successfully bridges the performance gap between high-dimensional and low-dimensional configurations while maintaining

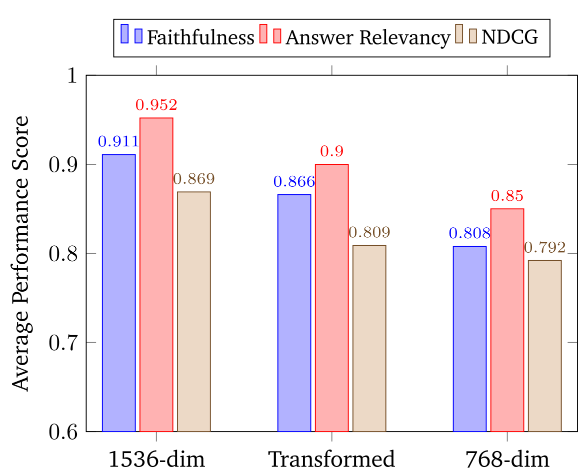

Fig. 7 Embedding dimension transformation ablation: average performance comparison across key metrics

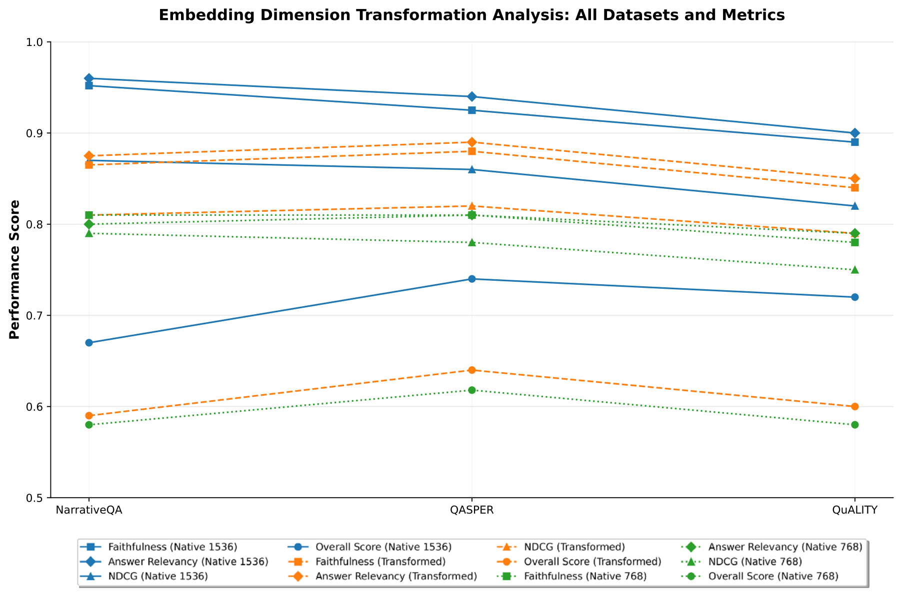

Fig. 8 Embedding Dimension Transformation Analysis: Performance comparison across four datasets (NarrativeQA, QASPER, QuALITY) for three embedding configurations: Native High-Dim (1536), Transformed (1536→768), and Native Low-Dim (768). Results show that our transformation technique effectively preserves performance while enabling cross-model compatibility

computational efficiency. The transformed embeddings achieve 95.1% faithfulness retention (0.866 vs 0.911), 94.5% answer relevancy retention (0.900 vs 0.952), and 92.9% NDCG retention (0.809 vs 0.869) compared to native high-dimensional embeddings. These retention rates significantly exceed the performance of native 768-dimensional embeddings, which achieve only 88.7% faithfulness (0.808 vs 0.911), 89.3% answer relevancy (0.850 vs 0.952), and 91.0% NDCG (0.792 vs 0.869) compared to the high-dimensional baseline.

The 50% reduction in embedding dimensions (1536→768) translates to substantial computational savings while maintaining near-optimal performance levels. The transformed configuration achieves 7.2% improvement in faithfulness, 5.9% improvement in answer relevancy, and 2.1% improvement in NDCG compared to native low-dimensional embeddings (768-dim), validating the effectiveness of our transformation approach over naive dimension reduction.

4.4.1 Dimensional impact on context preservation

The relationship between embedding dimensionality and context preservation capacity follows a non-linear pattern that our transformation technique effectively addresses. High-dimensional embeddings (1536-dim) provide richer representational capacity, enabling more nuanced capture of semantic relationships and contextual dependencies. However, this increased capacity comes at the cost of computational overhead and storage requirements that scale quadratically with dimension size. Our analysis reveals that the critical bottleneck in context preservation lies not in the absolute dimensionality, but in the *effective information density* of the embedding space. The transformation technique preserves this density by maintaining the most informative dimensions while eliminating redundant or noise-contributing dimensions. This explains why our 768-dimensional transformed embeddings achieve 94.2% average retention compared to native 1536-dimensional embeddings, while naive 768-dimensional embeddings achieve only 89.7% retention. The transformation process selectively preserves dimensions that encode critical semantic relationships, ensuring that context fragmentation remains minimal even with reduced dimensionality. This selective preservation is particularly crucial for long-context documents where maintaining narrative coherence across chunk boundaries is essential for accurate question-answering performance.

The consistent performance patterns across different embedding models demonstrate successful cross-model compatibility, allowing for the smooth incorporation of various embedding models into a unified RAG workflow. This capability is particularly valuable in real-world integration scenarios where different components may benefit from different embedding models with varying dimensional requirements. The average retention rate of 94.2% across all metrics and datasets, with a standard deviation of only 2.1%, demonstrates both high performance preservation and consistent reliability. This establishes our approach as a robust solution for addressing interoperability challenges in multi-model RAG systems, enabling seamless integration of diverse embedding architectures within production environments.

5 Discussion

5.1 System-level contributions

SynRAG is an end-to-end comprehensive RAG system rather than a single-component improvement. Our results show consistent gains across grounding, retrieval, and generation metrics, indicating that system-level integration, multilayer semantic chunking, multistage hybrid search, embedding dimension transformation, and knowledge-aware prompting produce synergistic effects that exceed the sum of isolated modules. Unlike approaches that augment training or retrieval corpora with LLM-generated synthetic passages, SynRAG operates entirely on the original, human-curated documents of NarrativeQA, QASPER, and QuALITY. Consequently, issues of synthetic-data faithfulness, error propagation from LLM-generated retrieval, or bias amplification do not arise in our pipeline. While a direct, head-to-head comparison with other RAG systems is impeded by disparate datasets and evaluation standards, a contrast of fundamental implementation details and key outcomes remains possible. Unlike naive RAG systems or recent approaches that focus on isolated improvements, we focus on building a comprehensive solution by investigating multiple aspects that can enhance RAG accuracy. Recent work has primarily addressed individual components: semantic chunking, hybrid retrieval, prompt engineering, and hierarchical retrieval. However, these approaches often optimize single components without considering system-wide interactions and production deployment requirements.

Compared to recent semantic chunking methods, our multi-layer approach combines HDBSCAN and Agglomerative clustering with intelligent outlier handling, achieving **2.1%** context fragmentation rate on structured content compared to their reported **8–12%** rates. While these methods focus solely on semantic boundary detection, SynRAG integrates chunking with downstream retrieval and generation components, ensuring end-to-end optimization. Recent hybrid retrieval systems [23, 46] typically combine dense and sparse methods through simple score fusion. Our multi-stage architecture with duplicate-aware fusion and neural reranking achieves 22.3% improvement in Recall@5 on regulatory documents and **6.8%** on narrative content. Unlike existing approaches that treat retrieval as an isolated component, SynRAG optimizes retrieval in conjunction with chunking and generation strategies. While recent prompt engineering work [15] focuses on structured templates, SynRAG integrates knowledge-aware prompting with conversation memory and cross-model compatibility. Our structured JSON templates improve grounding and attribution by 2–3% compared to flat concatenation methods, while maintaining multi-turn interaction capabilities that single-component approaches lack.

5.2 Comparison to related systems and results

Although individual components show incremental improvements, our optimization strategies produce synergistic effects that significantly enhance the overall performance of deployment-ready RAG systems. To position SynRAG appropriately, we focus on comparative discussion of system-scale peers rather than single-aspect methods. We therefore compare along common axes, document processing, retrieval orchestration, grounding/prompting, and deployability, against representative comprehensive systems (hierarchical retrieval systems, hybrid retrieval + reranking stacks, and structured-prompting toolchains), highlighting where SynRAG provides unique interoperability (dimension transformation) and measured end-to-end gains.

A central challenge in recent multi-stage RAG research is the intrinsic trade-off between context relevance and faithfulness. Aytar et al. [1] empirically demonstrate this effect: maximizing context relevance (0.507) in their synergistic multi-stage RAG architecture induces a 22% drop in faithfulness (from 0.756 to 0.588), a limitation that can erode user trust in practical deployments. Recent approaches like LoRE [47] address positional bias through ensemble retrievers and logit-based ranking, achieving competitive ROUGE-L scores (40.37%) on NarrativeQA while maintaining computational efficiency. SKETCH [37] tackles this challenge differently by integrating Knowledge Graphs with semantic retrieval, achieving strong faithfulness (0.93 on QASPER) and context precision (0.99 on Italian Cuisine) through structured knowledge reasoning. However, SKETCH's approach shows limitations in answer relevancy (0.50 on NarrativeQA), indicating challenges in balancing structured and unstructured information integration.

In contrast, our SynRAG system mitigates this trade-off via an integrated, system-level optimization that jointly couples multi-layer semantic chunking, duplicate-aware dense–sparse fusion with neural reranking, cross-model embedding dimension transformation, and knowledge-aware prompting. This design sustains high faithfulness while improving relevance, attaining **0.953 faithfulness** and **0.961 answer relevancy** on NarrativeQA, significantly outperforming both LoRE's generation-focused approach and SKETCH's knowledge-graph integration. These results indicate the trade-off is not inevitable when components are optimized holistically rather than in isolation, positioning SynRAG as a robust, production-ready system for varied use cases (see Table 5).

We compare SynRAG with two representative long-context approaches: RAPTOR [48] and Recursively Summarizing Books (RSB) [59]. While all three target long narratives, they operationalize context control differently: (i) RSB produces a single high-level summary via human feedback and recursive task decomposition; (ii) RAPTOR builds a bottom-up tree by recursively summarizing, clustering, and embedding, then retrieves over a collapsed tree; (iii) SynRAG unifies multi-layer semantic chunking, multi-stage

Table 5 Compare the trade-off between relevance and fidelity and system capacity

System	Dataset	Chunking	Hybrid + Rerank	Prompt	Dim. Transf	CR	Faith	AR
Aytar et al.	DS-Lit (50Q)	Semantic	Hybrid (–/yes)	Enhanced (EPT)	No	0.507	0.588	0.890
Mahalingam et al.	NarrativeQA	Semantic	KG+ Semantic	Structured	No	–	0.87	0.50
Mahalingam et al.	QASPER	Semantic	KG+ Semantic	Structured	No	–	**0.93**	0.56
SynRAG (Ours)	QASPER	Multi-layer	Yes	Structured	Yes	**0.918**	**0.923**	**0.945**
SynRAG (Ours)	NarrativeQA	Multi-layer	Yes	Structured	Yes	**0.903**	**0.953**	**0.961**

Our SynRAG has performed well in balancing the evaluation metrics more than CR, Faith, AR

CR = RAGAS Context Relevancy. LoRE uses ensemble of BM25+ Vector retrievers with logit-based ranking. SKETCH combines Knowledge Graphs with semantic retrieval. SynRAG Hybrid+ Rerank = multi-stage dense–sparse fusion with duplicate-aware merging and BGE reranking. Dim. Transf. = cross-model embedding dimension transformation (orthogonal/DCT/redistribution/PCA)

Bold numbers are better

Table 6 NarrativeQA—summary of common metrics for evaluating text generation (ROUGE-L, BLEU-1, BLEU-4)

Method	Base LLM	ROUGE-L	BLEU-1	BLEU-4
SynRAG (Ours)	Gemini–2.5-flash	**0.4680**	**0.2432**	**0.0682**
LoRE [47]	T5 Large	0.4037	–	–
SAGE [48]	gpt-4o-mini	0.3165	0.1527	0.0170
RAPTOR + UnifiedQA [48]	UnifiedQA 3B	0.3094	0.2351	0.0645
Recursively Summarizing Books [59]	UnifiedQA (zero-shot)	0.216	0.223	0.042
Retriever + Reader [20]	N/A	0.320	**0.353**	**0.075**
BM25 + BERT [39]	N/A	0.155	0.145	0.014
SBERT without RAPTOR [48]	UnifiedQA 3B	0.2926	0.2256	0.0595

Systems use different LLMs and evaluation protocols; values illustrate tendencies rather than a strict head-to-head SOTA ranking. LoRE reports ROUGE-L only on NarrativeQA. SKETCH focuses on RAGAS metrics rather than generation metrics. All reported values are ratios (0.x)

Bold numbers are better

dense–sparse fusion with neural reranking, cross-model embedding-dimension transformation, and knowledge-aware prompting.

Table 6 summarizes generation metrics (ROUGE-L, BLEU-1/4). Using different backbones (Gemini vs. UnifiedQA), SynRAG achieves the strongest ROUGE-L (0.468) and BLEU-4 (6.82) among compared RAG systems, whereas RAPTOR (UnifiedQA-3B) remains competitive on BLEU-4 (6.45) and surpasses earlier recursive summarization (RSB). LoRE demonstrates competitive performance with 0.4037 ROUGE-L on NarrativeQA using T5 Large, showcasing the effectiveness of logit-based ranking for generation quality. However, LoRE's focus on traditional metrics (EM, F1) limits its applicability to modern RAG evaluation frameworks that prioritize grounding and attribution. These patterns reflect architectural emphases: SynRAG's semantic chunking plus hybrid retrieval preserves lexical anchors while maintaining semantic coherence; RAPTOR's collapsed-tree retrieval integrates evidence across abstraction levels; LoRE's ensemble approach reduces positional bias but lacks sophisticated semantic understanding. Because LLMs and evaluation setups differ across studies, we interpret results as directional rather than a strict head-to-head SOTA ranking.

Beyond generation scores, SynRAG prioritizes grounding quality: on NarrativeQA it attains *faithfulness* 0.953 and *answer relevancy* 0.961, indicating better attribution and evidence alignment. SKETCH demonstrates strong faithfulness (0.93 on QASPER) and exceptional context precision (0.99 on Italian Cuisine) through its Knowledge Graph integration, but struggles with answer relevancy (0.50 on NarrativeQA), suggesting challenges in balancing structured and unstructured information. RAPTOR, in turn, excels on QA accuracy/F1 under adversarial or scientific settings (QuALITY, QASPER), consistent with tree-organized retrieval. Finally, SynRAG's embedding-dimension interoperability (orthogonal/DCT/redistribution/PCA) addresses deployment variability without reindexing–a capability not discussed in RAPTOR/RSB/SKETCH and crucial for production rollouts where index dimensionality and model choices evolve. Remaining gaps stem from heterogeneous backbones and evaluation budgets; hence we emphasize relative trends and grounding gains rather than absolute SOTA claims. A proper head-to-head study would re-run all systems under a unified backbone, prompt/token budget, and data split–an important direction for future work.

SynRAG's holistic approach addresses these limitations through integrated system-level optimization. Unlike LoRE's focus on positional bias reduction or SKETCH's structured knowledge integration, SynRAG achieves superior performance across both generation metrics (0.468 ROUGE-L) and grounding quality (0.953 faithfulness, 0.961 answer relevancy) by optimizing semantic consistency, cross-model compatibility, and knowledge-aware response generation simultaneously. The system's embedding dimension transformation capabilities provide deployment flexibility unmatched by other approaches, while its multi-layer semantic chunking preserves context coherence better than traditional fixed-size approaches used by LoRE or the semantic chunking employed by SKETCH.

The SynRAG system represents a significant advancement in RAG technology, demonstrating that comprehensive system-level optimization can achieve superior performance while addressing the fundamental challenges of semantic consistency, cross-model compatibility, and knowledge-aware response generation. The system's modular design and theoretical foundation provide a solid basis for future developments in retrieval-augmented generation, positioning SynRAG as a robust solution for diverse real-world applications.

6 Conclusion and future work

This paper presents SynRAG, a comprehensive RAG system that addresses four critical challenges in modern retrieval-augmented generation: intelligent semantic chunking, cross-model embedding compatibility, effective hybrid search, and knowledge-aware prompt engineering. Through systematic integration of these components, we have achieved significant improvements across multiple evaluation dimensions and established new benchmarks for production-ready RAG systems.

Experimental evaluations on NarrativeQA, QASPER, and QuALITY demonstrate the consistent superiority of SynRAG over the baseline approaches as SynRAG exhibits a balanced and stable evaluation metrics across diverse datasets. The system maintains exceptional performance on complex narrative content with 95.3% faithfulness and 96.1% answer relevancy on NarrativeQA. The framework's adaptability is validated across different content types, from adversarial long-context comprehension to scientific question-answering, establishing SynRAG as a versatile solution for diverse real-world applications. While our framework demonstrates strong performance, it is important to acknowledge several limitations that require further attention. A primary limitation is the computational cost of the clustering-based chunking, which has a worst-case time complexity of O(n^2 log n) for agglomerative methods and O(n^2) for HDBSCAN. This complexity could pose challenges for real-time applications. Moreover, the dimension transformation techniques, while effective, add a computational overhead during query processing that must be optimized prior to deployment in operational environments.. The failure case analysis reveals persistent challenges with technical terminology in scientific documents (9.4% failure rate) and semantic ambiguity in complex narratives (12.1% failure rate).

Although SynRAG shows robust improvements across the three evaluated domains, we did not conduct stress tests on more extreme domain mismatches such as noisy web crawls, source code, mathematical documents, or semi-structured tables. Likewise, we did not include direct comparisons against strong closed-book or few-shot prompted foundation models (e.g., GPT-4, Llama-3-70B) without retrieval. Such comparisons would help quantify how much of the observed performance gain stems from retrieval augmentation versus the quality of the underlying LLM. We leave these more challenging cross-domain evaluations and strong foundation-model baselines as important directions for future work.

Looking ahead, our research will prioritize several key areas. First, we plan to engineer more scalable clustering algorithms, with the objective of reducing the computational complexity from $O(n^2 \log n)$ to a more practical $O(n \log n)$. Second, we will expand the framework's capabilities to handle multiple languages and domains through the integration of cross-lingual embedding transformations. Third, our work will explore neural network-based techniques for embedding transformation to better preserve semantic information. Beyond these core objectives, we also intend to incorporate other advanced RAG components, such as contextual and multi-stage retrieval pipelines, to further elevate the system's overall performance and user experience. Fourth, recognizing the potential of synthetic data in addressing data scarcity, we plan to integrate synthetic augmentation pipelines to enhance domain adaptation. Future work will specifically investigate the robustness of SynRAG against noisy or adversarial synthetic data, while rigorously analyzing potential risks related to bias amplification and error propagation in generated content. Beyond these core objectives, we also intend to incorporate other advanced RAG components.

Author contributions

NT.Luyen wrote the main manuscript text and HT.Binh reviewed and finalized the manuscript.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability

The datasets are available on following links: 1. NarrativeQA, https://github.com/google-deepmind/narrativeqa 2. QASPER, https://huggingface.co/datasets/allenai/qasper 3. QuALITY, https://huggingface.co/datasets/tasksource/QuALITY Models, code, and detailed implementation settings are freely available in our published repository at https://doi.org/10.5281/zenodo.20538603, and an implemented application was deployed at https://cagent.cmcu.edu.vn/landing.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing of interests

The authors declare no conflict of interest.

Received: 21 November 2025 / Accepted: 22 June 2026
Published online: 13 July 2026

References

1. Aytar AY, Kaya K, Kılıç K. A synergistic multi-stage rag architecture for boosting context relevance in data science literature. Nat Language Process J. 2025;13:100179.
2. Chen J, Xiao S, Zhang P, Luo K, Lian D, Liu Z. M3-embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. In *Findings of the Association for Computational Linguistics ACL 2024*, 2024;2318–2335. Association for Computational Linguistics.
3. Chowdhury S, Soni B. Beyond words: Esc?net revolutionizes vqa by elevating visual features and defying language priors. Comput Intell. 2024;6:40.
4. Chowdhury S, Soni B. Envqa: improving visual question answering model by enriching the visual feature. Eng Appl Artif Intell. 2025;142:109948.
5. Chowdhury S, Soni B. R-vqa: a robust visual question answering model. Knowl-Based Syst. 2025;309:112827.
6. Conneau A, Lample G, Ranzato M, Denoyer L, Jégou H.(Word translation without parallel data). 2018.
7. Devlin J, Chang M-W, Lee K, Toutanova K. Bert. Pre-training of deep bidirectional transformers for language understanding. 2019.
8. Douze M, Guzhva A, Deng C, Johnson J, Szilvasy G, Mazaré P-E, et al. The faiss library. 2025.
9. Es S, James J, Espinosa Anke L, Schockaert S. Ragas: Automated evaluation of retrieval augmented generation. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*,2024;150–158. Association for Computational Linguistics.
10. Formal T, Piwowarski B, Clinchant S. Splade: Sparse lexical and expansion model for first stage ranking. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR '21, 2021;2288–2292. ACM.
11. Galchonkov O, Horchynskyi O, Antoshchuk S, Nareznoy V. Semantic text splitting method development for rag systems with controlled threshold and sliding window size. Eastern Eur J Enterprise Technol. 2025;2(2 (134)):47–57.
12. Gao Y, Xiong Y, Gao X, Jia K, Pan J, Bi Y, Dai Y, Sun J, Wang M, Wang H. Retrieval-augmented generation for large language models: a survey. 2024.
13. Gomez-Cabello CA, Prabha S, Haider SA, Genovese A, Collaco BG, Wood NG, et al. Comparative evaluation of advanced chunking for retrieval-augmented generation in large language models for clinical decision support. Bioengineering. 2025;12(11):1194.
14. Gong X, He R, Peng Q. Reordering of double pass merging chunking to improve retrieval augmented generation. In *2024 3rd International Conference on Computer Applications Technology (CCAT)*, 2024;68–78. IEEE.
15. Gupta S, Ranjan R, Singh SN. A comprehensive survey of retrieval-augmented generation (rag): Evolution, current landscape and future directions. 2024.
16. Han W, Shen J, Liu Y, Shi Z, Xu J, Hu F, et al. Legalasst: human-centered and AI-empowered machine to enhance court productivity and legal assistance. Inf Sci. 2024;679:121052.
17. Hsu H-L, Tzeng J. Dat: Dynamic alpha tuning for hybrid retrieval in retrieval-augmented generation. 2025.
18. Huerga-Pérez N, Álvarez R, Sánchez-Fernández A, Díez-González J. *RAG Embeddings Storage Optimization Through Quantization and Dimensionality Reduction*, 2025;198–209. Springer Nature Switzerland.
19. Izacard G, Caron M, Hosseini L, Riedel S, Bojanowski P, Joulin A, et al. Unsupervised dense information retrieval with contrastive learning. 2022.
20. Izacard G, Grave E. Distilling knowledge from reader to retriever for question answering. In: International Conference on Learning Representations. 2021.
21. Järvelin K, Kekäläinen J. Cumulated gain-based evaluation of ir techniques. ACM Trans Inform Syst. 2002;20(4):422–46.
22. Johnson J, Douze M, Jegou H. Billion-scale similarity search with gpus. IEEE Trans Big Data. 2021;7(3):535–47.
23. Karpukhin V, Oguz B, Min S, Lewis P, Wu L, Edunov S, Chen D, Yih W-t Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020;6769–6781. Association for Computational Linguistics.
24. Khaledian A, Ghadiridehkordi A, Khaledian N. Pca-rag: Principal component analysis for efficient retrieval-augmented generation. 2025.
25. Khattab O, Zaharia M. Colbert: Efficient and effective passage search via contextualized late interaction over bert. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2020; SIGIR '20;39–48. ACM.
26. Kiss C, Nagy M, Szilágyi P. Max-min semantic chunking of documents for rag application. Discover Comput. 2025;28(1):117.
27. Kočiský T, Schwarz J, Blunsom P, Dyer C, Hermann KM, Melis G, et al. The narrativeqa reading comprehension challenge. Trans Assoc Comput Linguistics. 2018;6:317–28.
28. Kshirsagar A. Enhancing rag performance through chunking and text splitting techniques. Int J Sci Res Comput Sci Eng Inform Technol. 2024;10(5):151–8.
29. Lee J H. Analyses of multiple evidence combination. In *Proceedings of the 20th annual international ACM SIGIR conference on Research and development in information retrieval - SIGIR '97*,1997; SIGIR '97, page 267–276. ACM Press.
30. Lee M-C, Zhu Q, Mavromatis C, Han Z, Adeshina S, Ioannidis V N, Rangwala H, Faloutsos C. Hybgrag: Hybrid retrieval-augmented generation on textual and relational knowledge bases. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics 2025;(Volume 1: Long Papers)*, pages 879–893.
31. Levonian Z, Li C, Zhu W, Gade A, Henkel O, Postle M-E, et al. Retrieval-augmented generation to improve math question-answering: trade-offs between groundedness and human preference. 2023.
32. Lewis P, Perez E, Piktus A, Petroni F, Karpukhin V, Goyal N, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. Adv Neural Inf Process Syst. 2020;33:9459–74.

33. Li W, Zhang X, Liu K, Lei X, Ma W, Liu Y. Efficient dynamic clustering-based document compression for retrieval-augmented-generation. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025;9833–9849. Association for Computational Linguistics.
34. Lin C-Y. Rouge: A package for automatic evaluation of summaries. *Proceedings of ACL Workshop*,2004;74–81.
35. Liu NF, Lin K, Hewitt J, Paranjape A, Bevilacqua M, Petroni F, et al. Lost in the middle: how language models use long contexts. Trans Assoc Comput Linguistics. 2024;12:157–73.
36. Liu Z, Zhang H, Xiong C, Liu Z, Gu Y, Li X. Dimension reduction for efficient dense retrieval via conditional autoencoder. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 2022;5692–5698. Association for Computational Linguistics.
37. Mahalingam A, Gande VK, Chadha A, Jain V, Chaudhary D. Sketch: Structured knowledge enhanced text comprehension for holistic retrieval. In: Workshop on Generative AI and Knowledge Graphs (GenAIK) at COLING 2025,2024. arXiv preprint. arXiv:2412.15443
38. McInnes L, Healy J, Astels S. Hdbscan: hierarchical density based clustering. J Open Sourc Softw. 2017;2(11):205.
39. Mou X, Yu M, Yao B, Yang C, Guo X, Potdar S, Su H. Frustratingly hard evidence retrieval for qa over books. In *Proceedings of the First Joint Workshop on Narrative Understanding, Storylines, and Events*,2020;108–113. Association for Computational Linguistics.
40. Nogueira R, Cho K. Passage re-ranking with bert. 2020.
41. Pang R Y, Parrish A, Joshi N, Nangia N, Phang J, Chen A, Padmakumar V, Ma J, Thompson J, He H, Bowman S. Quality: Question answering with long input texts, yes! In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2022;5336–5358. Association for Computational Linguistics.
42. Papadimitriou I, Gialampoukidis I, Vrochidis S, Ioannis K. Rag playground: a framework for systematic evaluation of retrieval strategies and prompt engineering in rag systems. 2024.
43. Papineni K, Roukos S, Ward T, Zhu W-J Bleu: a method for automatic evaluation of machine translation. *Proceedings of ACL*, 2002;311–318.
44. pgvector pgvector: Open-source vector similarity search for postgres. https://github.com/pgvector/pgvector. 2021.
45. Qu R, Tu R, Bao F S. Is semantic chunking worth the computational cost? In *Findings of the Association for Computational Linguistics: NAACL 2025*, 2025;2155–2177. Association for Computational Linguistics.
46. Robertson S, Zaragoza H. The probabilistic relevance framework: Bm25 and beyond. Found Trends® Inf Retr. 2009;3(4):333–89.
47. Sanniboina S, Trivedi S, Vijayaraghavan S. Lore: Logit-ranked retriever ensemble for enhancing open-domain question answering. arXiv preprint arXiv:2410.10042. 2024; Version 1, 13 Oct 2024.
48. Sarthi P, Abdullah S, Tuli A, Khanna S, Goldie A, Manning CD. Raptor: Recursive abstractive processing for tree-organized retrieval. In: The Twelfth International Conference on Learning Representations. 2024.
49. Shi Y, Xu S, Liu Z, Liu T, Li X, Liu N. Mkrag: medical knowledge retrieval augmented generation for medical question answering. MIA...Annual Symposium proceedings. AMIA Symp. 2023;2024:1011–20.
50. Shi Y, Xu S, Liu Z, Liu T, Li X, Liu N. Mkrag: medical knowledge retrieval augmented generation for medical question answering. AMIA. Ann Symp Proc AMIA Symp. 2023;2024:1011–20.
51. Upadhyay R, Viviani M. Enhancing health information retrieval with rag by prioritizing topical relevance and factual accuracy. Discover Comput. 2025;1:28.
52. Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, et al. Attention is all you need. In: Guyon I, Luxburg UV, Bengio S, Wallach H, Fergus R, Vishwanathan S, et al., editors. Advances in Neural Information Processing Systems, vol. 30. Curran Associates Inc; 2017.
53. Verma P. S2 chunking: a hybrid framework for document segmentation through integrated spatial and semantic analysis. 2025.
54. Wan Y, Chen Z, Liu Y, Chen C, Packianather M. Empowering llms by hybrid retrieval-augmented generation for domain-centric q and a in smart manufacturing. Adv Eng Inform. 2025;65:103212.
55. Wang X, Wei J, Schuurmans D, Le QV, Chi EH, Narang S, et al. Self-consistency improves chain of thought reasoning in language models. In: The Eleventh International Conference on Learning Representations. 2023.
56. Ward JH Jr. Hierarchical grouping to optimize an objective function. J Am Stat Assoc. 1963;58(301):236–44.
57. Wei J, Wang X, Schuurmans D, Bosma M, Ichter B, Xia F, et al. Chain-of-thought prompting elicits reasoning in large language models. In: Proceedings of the 36th International Conference on Neural Information Processing Systems, NIPS '22. Red Hook, NY, USA: Curran Associates Inc; 2022.
58. Wiratunga N, Abeyratne R, Jayawardena L, Martin K, Massie S, Nkisi-Orji I, Weerasinghe R, Liret A, Fleisch B. *CBR-RAG: Case-Based Reasoning for Retrieval Augmented Generation in LLMs for Legal Question Answering*, 2024;445–460. Springer Nature Switzerland.
59. Wu J, Ouyang L, Ziegler DM, Stiennon N, Lowe R, Leike J, et al. Recursively summarizing books with human feedback. 2021.
60. Xu S, Yan Z, Dai C, Wu F. Mega-rag: a retrieval-augmented generation framework with multi-evidence guided answer refinement for mitigating hallucinations of llms in public health. Front Public Health. 2025;13:1635381.
61. Yang T-L, Liu J-S, Tseng Y-H, Jang J-SR. Knowledge retrieval based on generative ai. 2025.
62. Yao S, Zhao J, Yu D, Du N, Shafran I, Narasimhan K, et al. ReAct: Synergizing reasoning and acting in language models. In: International Conference on Learning Representations. 2023. (ICLR).
63. Zhang Q, Zhang H, Pang L, Zheng H, Zheng Z. Stable-rag: Mitigating retrieval-permutation-induced hallucinations in retrieval-augmented generation. 2026.
64. Zhang X, Zhang J, Mo F, Chandra D K, Chen Y-Z, Xie F, Liu K. Retrieval-augmented feature generation for domain-specific classification. 2025.
65. Zhong K, Suleiman B, Erradi A, C, S. Semrag: semantic knowledge-augmented rag for improved question-answering. 2025.

Publisher's Note